SUNY Old Westbury 2026-2027 Undergraduate Catalog

Course Descriptions

Contract All Courses |

Chemistry, Physics and Physical Science

CP 2000 - Chemistry for Non-majors

Credits: 3

Survey of chemistry for the non-science major. Basic principles to be covered include atomic structure, chemical properties, chemical reactions, the nature of organic and inorganic compounds and novel materials. Principles are illustrated with applications to daily life. No prior background in chemistry required.

Prerequisites

MA 1010

, completed with grade of "C" or better

When Offered:

Periodically.

CP 2001 - Chemistry for Non-Majors Laboratory

Credits: 1

Laboratory course designed to illustrate concepts taught in Chemistry for Non-majors. Emphasis on measurement, observation, data collection and reporting in addition to illustration of basic chemical principles. One three-hour laboratory session per week.

When Offered:

In conjunction with CP 2000

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CP 2025 - Renewable Energy Systems

Credits: 4

General Elective science course illuminating the different forms of renewable energy and the scope within different sectors for a wide range of majors.

When Offered:

Every Semester

CP 2050 - The Nature and Development of Science

Credits: 4

The course involves students in the study and reflection within the historical context of the development of science as a forum to synthesize their learning of various scientific disciplines. The context is provided by a thorough analysis of the historical and philosophical foundations of scientific knowledge and practice, and its emergency as a distinctly modern view of the universe. The course addresses National Science Education Standards that call for an understanding of the history and development of science, as well as its interdisciplinary nature. Note: This course does not fulfill the Liberal Education Curriculum Domain Science requirement.

Prerequisites

EMS

CP 2070 - Introduction to Forensic Science

Credits: 4

This course is an introduction to forensic chemistry intended for non-science majors. The course examines the many ways that chemical principles and scientific techniques are applied in modern crime investigation: the chemical characteristics of fuels and explosives as well as arson-related methods; other ways of identifying criminals through chemical processes, including blood and bodily fluid analyses, the chemical characteristics found in paints, fibers, hair, DNA, and fingerprinting; and the chemical properties of drug analysis. Students learn the fundamentals of forensic science and applications in crime investigation and adjudication.

When Offered:

Every Semester

CP 2120 - Principles of Chemistry I

Credits: 3

Introduction to the origins and principles of modern chemistry. Discussion of atomic and molecular structure, stoichiometry, periodicity, bonding and states of matter. Principles are illustrated through demonstrations and study of descriptive chemistry.

Prerequisites

MA 1020, completed with grade "C" or better

When Offered:

Every semester.

CP 2121 - Principles of Chemistry Lab I

Credits: 1

Laboratory course designed to illustrate concepts of Principles of Chemistry I. Emphasis on basic chemical concepts, quantitative laboratory skills, descriptive chemistry, and development of scientific report writing skills. Three laboratory hours per week.

When Offered:

In conjunction with CP 2120.

CP 2130 - Principles of Chemistry II

Credits: 3

Continuation of Principles of Chemistry I. Discussion of acid/base and oxidation/reduction reactions, equilibrium, kinetics and electrochemistry. Principles are illustrated through demonstrations and study of descriptive chemistry.

Prerequisites

CP 2120 completed with grade of "C" or better

When Offered:

Every semester.

CP 2131 - Principles of Chemistry Lab II

Credits: 1

Continuation of CP 2121. Emphasis on basic chemical concepts, quantitative laboratory skills, descriptive chemistry, and development of scientific report writing skills. Three laboratory hours per week.

Prerequisites

CP 2121 completed with grade of "C" or better;

When Offered:

In conjunction with CP 2130.

CP 2210 - Physical Science

Credits: 3

A survey of basic concepts of physical science for students in non-science majors. Topics chosen from fundamental principles of physics, astronomy, chemistry and earth science will provide the student with a broad basic background.

Prerequisites

MA 1020

, completed with grade of "C" or better

When Offered:

Periodically.

CP 2211 - Physical Science Laboratory

Credits: 1

A one semester hands-on laboratory course designed to introduce the non-science major to the basic concepts of the physical sciences. Laboratory exercises are selected to provide a broad background in the fundamental aspects of

Mechanics, Light, Electricity, Chemistry, Geology and Astronomy.

Prerequisites

MA 1020

, completed with grade of “C” or better;

When Offered:

In conjunction with CP 2210

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CP 2300 - Weather Studies

Credits: 3

Weather Studies covers the composition and structure of the atmosphere; the flows of energy to, from, and through the atmosphere; and the resulting atmospheric motions. The physical principles of atmospheric phenomena are stressed, as well as the effects of weather on society. Students will participate in learning activities coordinated with current weather data delivered via the web. Note: This is an online course; concurrent registration in CP 2301

required.

Prerequisites

Math Proficiency;

When Offered:

Periodically.

CP 2301 - Weather Studies Lab

Credits: 1

Laboratory course designed to illustrate concepts taught in Weather Studies (CP 2300

). Laboratory investigations include experiments on the physical properties of air, analysis of weather trends, and creation and interpretation of real time weather plots. Note that all lab work will be done online.

When Offered:

In conjunction with CP 2300

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CP 2400 - Ocean Studies

Credits: 3

This is an introductory oceanography course suitable for students with no prior science background. This course investigates the world's oceans from an earth system perspective, emphasizing interactions between spheres (i.e., hydrosphere, atmosphere). Topics covered will include the physical and chemical properties of seawater, flow and transformations of water and energy, and ocean circulation. Students will also study adaptations of marine life and human societal impacts on the oceans. Learning activities will make use of real-time ocean data accessible via the web. Note: This course is conducted online.

Prerequisites

Math Proficiency;

CP 2401 - Ocean Studies Lab

Credits: 1

Laboratory course designed to illustrate concepts taught in Ocean Studies (CP 2400

). Laboratory investigations will be both hands-on and web-based, and will include the interpretation of real-time ocean data.

When Offered:

In conjunction with CP 2400

. Successful completion of Ocean Studies and Ocean Studies Lab satisfies the college Liberal Education Curriculum requirement for a laboratory science course.

CP 2500 - Climate Studies

Credits: 3

An introductory course in climate studies that uses current and historical climate data available on the internet to explore climate change and its potential impacts. In order to evaluate the emerging evidence about climate change,

individuals need to understand the factors that affect climate and its variability. Major earth, air and water systems, and their relationship to climate and to climate change, are discussed. Emphasis is placed on exploring the tools scientists use to understand and describe climate, including satellite observations, ice cores and mathematical modeling. The public policy aspects and societal impacts of climate change will be debated. Taught in a hybrid mode, the course includes activities such as in-class presentations, exams and the acquisition, analysis and presentation of online climate data.

Prerequisites

Math Proficiency;

CP 2501 - Climate Studies Lab

Credits: 1

Laboratory course offered in conjunction with, and designed to illustrate, concepts taught in Climate Studies (CP 2500). Laboratory investigations will be web-based, and will include analysis and interpretation of climate data.

CP 2600 - Environmental Analysis of Long Island

Credits: 4

This is an introductory course that will focus on the study of the natural sciences of Long Island. It includes many aspects of biology, earth and atmospheric sciences, fundamental principles of chemistry and physics, human population dynamics, and an appreciation for biological and natural resources. It will always include the consideration of people and how they have influenced the system under examination. This course will use laboratory, field and a case study approach to give the students a “hands-on” approach to the fundamental principles of environmental analysis. There will be an emphasis on laboratory and field methodologies, data analysis, and technical communication within an interdisciplinary context.

Prerequisites

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CP 2800 - Nutrition and Health

Credits: 4

An introductory nutrition course suitable for non-science majors. Includes application of nutrition principles to daily dietary practice. Macro & micronutrients, energy metabolism and balance, weight management and fitness, life cycle nutrition at different stages of life, and food and the global environment. Major chronic diseases linked to nutrition and current topics in nutrition will be discussed. Students will complete detailed individual analysis of their diet and will gain the knowledge and skills needed to achieve a nutritionally healthy lifestyle. Course includes laboratory exercises.

Prerequisites

Math Proficiency;

CP 3230 - Mathematical Methods in the Physical Sciences

Credits: 3

This course covers a variety of essential mathematical concepts and techniques used in chemistry, focusing on applications to physical problems rather than on deviations and proofs. Topics will include infinite series, complex functions, differential equations, vector spaces and linear algebra, and Fourier analysis.

Prerequisites

MA 2320

When Offered:

Fall

Note:

This course is crosslisted with PX 3100 Mathematical Methods in Physics

CP 3300 - Organic Chemistry I

Credits: 3

The first semester of an introductory organic chemistry course. Discussion of major classes of organic compounds, with emphasis on structure, reactions, synthesis, stereochemistry and reaction mechanism.

Prerequisites

CP 2130 completed with grade of "C" or better

When Offered:

Every Fall.

CP 3302 - Organic Chemistry Lab I

Credits: 2

Designed to illustrate topics of CP 3300. Stresses development of laboratory skills in synthesis, separation and purification of organic compounds. Emphasizes scientific report writing skills. Four laboratory hours per week.

Prerequisites

CP 2130 and CP 2131 with grades of "C" or better;

When Offered:

In conjunction with CP 3300.

CP 3310 - Organic Chemistry II

Credits: 3

Continuation of Organic Chemistry I, with discussion of additional major classes of organic compounds.

Prerequisites

CP 3300

When Offered:

Every Spring.

CP 3312 - Organic Chemistry Lab II

Credits: 2

Designed to illustrate topics of CP 3310. Stresses additional synthetic techniques not covered in CP 3302, spectroscopic methods and qualitative organic analysis. Emphasizes scientific report writing skills. Four laboratory hours per week.

Prerequisites

CP 3302

When Offered:

In conjunction with CP 3310.

CP 3400 - Analytical Chemistry

Credits: 5

Study of chemical stoichiometry, homogeneous and heterogeneous equilibria, principles of volumetric and gravimetric analysis, chemical separation methods and spectrophotometry. Includes statistical analysis of data and scientific record-keeping. Two lectures and one four-hour laboratory per week.

Prerequisites

CP 2130

, CP 2131

completed with grades of "C" or better

When Offered:

Every Fall.

CP 3450 - Inorganic Chemistry

Credits: 3

Study of inorganic chemistry emphasizing the use of thermodynamic, kinetic and quantum mechanical principles. Topics include structure, bonding, descriptive chemistry, coordination compounds, group theory, and discussion of current topics from the literature.

Prerequisites

CP 2130

When Offered:

Periodically.

CP 4320 - Advanced Organic Chemistry

Credits: 3

An examination of modern synthetic methods, molecular orbital theory, linear free energy relationships, acid-base catalysis, neighboring group participation, medium and salt effects, and a detailed consideration of the mechanistic features of a wide variety of organic reactions.

Prerequisites

CP 3310

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When Offered:

Periodically.

CP 4490 - Biochemistry for Life Sciences

Credits: 3

One semester survey course designed for life science majors requiring only one semester of biochemistry. We will cover basic concepts about pH, biological buffers, bioenergetics, nucleotides, amino acids, proteins, enzymes, carbohydrates, lipids, intermediary metabolism, and molecular biology. **This course is not open to Biochemistry majors for credit.**

Prerequisites

CP 3310 with a grade of C or better

CP 4510 - Biochemistry I

Credits: 3

Comprehensive coverage of the chemistry and metabolism of biological compounds, including proteins, nucleic acids, lipids, polysaccharides, and their precursors. Also treats enzyme kinetics, bioenergetics and biosynthesis of important biomolecules.

Prerequisites

CP 3310 with a grade of C or better

When Offered:

Every Fall.

CP 4515 - Biochemistry II

Credits: 3

Continuation of Biochemistry I, with emphasis on the control of biochemical processes, photosynthesis, replication, transcription and translation, genetic engineering, ribosomes, and metabolism.

Prerequisites

CP 4510

with a grade of C or better

When Offered:

Every Spring.

CP 4520 - Biochemistry Lab

Credits: 2

Laboratory experiments illustrating current biochemical techniques and reinforcing lecture topics. Four laboratory hours per week.

When Offered:

In conjunction with CP 4510.

CP 4600 - Forensic Chemistry Techniques

Credits: 3

This course offers an introduction to the role of chemistry in criminal investigations, with a focus on the principles, techniques, and methods used to analyze physical evidence in forensic science. Topics include the identification and

analysis of substances like drugs, toxins, and explosives, along with the use of tools such as chromatography and spectroscopy. Students will explore how forensic chemists aid in solving crimes. The course emphasizes the foundational principles of analytical chemistry and hands-on experience in forensic analysis and evidence handling. The prerequisite for CP4600 is successful completion of Analytical Chemistry (CP3400) lectures and labs with a grade of C or better. The CP4600 is completed over 7 weeks during Term A.

Prerequisites

Analytical Chemistry (CP3400) lectures and labs with a grade of C or better.

When Offered:

Spring Only

CP 4700 - Physical Chemistry I

Credits: 3

Survey of the basic principles of physical chemistry, with emphasis on thermodynamics, changes of state, solutions, phase equilibria and kinetic theory.

Prerequisites

CP 2130

completed with grade of "C" or better; MA 2320

completed with a grade of "C" or better, PX 2300

or PX 2100

completed with a grade of "C"or better

When Offered:

Every Fall

Note:

This course is crosslisted with PX 4100 Thermodynamics.

CP 4710 - Physical Chemistry II

Credits: 3

Survey of the basic principles of physical chemistry with emphasis on chemical kinetics, quantum mechanics, atomic and molecular structure.

Prerequisites

CP 2130

completed with a grade of "C" or better; MA 2320

completed with a grade of "C" or better, PX 2300

or PX 2100

completed with a grade of "C" or better

When Offered:

Spring

Note:

This course is cross-listed with PX 4300 Applied Quantum Mechanics

CP 4720 - Physical Chemistry Lab

Credits: 2

Laboratory experiments designed to illustrate the chemical principles underlying thermodynamics, equilibria, surface and transport properties, chemical kinetics, and spectroscopy. Includes vacuum techniques and use of computers to acquire and analyze data. Emphasizes scientific report writing in standard format.

When Offered:

In conjunction with CP 4710

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CP 4800 - Instrumental Analysis

Credits: 5

A study of organic and inorganic compounds using instrumental methods of characterization. Synthetic techniques may include use of inert atmosphere and electrochemical methods. Characterization of compounds may utilize spectroscopy, GC/MS, LC/MS, electrochemistry, NMR and others. Two lectures and one four-hour laboratory per week. Emphasis on scientific report writing in standard format.

Prerequisites

CP 3310

, CP 3312

, CP 3400

, with a grade of "C"or better

When Offered:

Every Spring.

CP 5500 - Advanced Topics in Chemistry

Credits: 3

One-semester lecture course covering selected topics of current importance to chemists. Course content may include such topics as spectroscopy, radiation chemistry, electricity and magnetism, molecular orbital theory, group theory and new instrumental techniques.

Prerequisites

Permission of the Instructor

When Offered:

Periodically.

CP 5900 - Research

Credits: 2-4

Research project carried out under the supervision of a faculty member. Each project includes a survey of the appropriate original literature, a theoretical or experimental investigation, and a comprehensive report of the results. Open only to students accepted as research trainees sponsored by a member of the Chemistry and Physics faculty.

Prerequisites

Permission of the Chemistry/Physics faculty

When Offered:

Every semester. Grading will be CR/NC.

CP 5920 - Seminar I

Credits: 1

Capstone course in which the student integrates coursework in the sub-disciplines to discuss topics of current interest in the profession. Includes instruction in library resources and on-line searching, scientific writing, ethics workshops and oral presentations. Students choose a supervisor and seminar topic and submit a draft written report. Attendance at Science Seminar Series is required.

Prerequisites

Senior Standing or Permission of the Instructor

When Offered:

Every Fall.

CP 5921 - Seminar II

Credits: 1

Continuation of CP 5920

. Emphasis on producing paper in American Chemical Society format. Attendance at Science Seminar Series is required.

Prerequisites

CP 5920

When Offered:

Every Spring.

CP 9990 - Independent Study in Chemistry/Physics

Credits: 2-4

An academic project other than laboratory research carried out under the supervision of a faculty member.

Prerequisites

Permission of the Chemistry/Physics faculty

When Offered:

Every semester.

Contract All Courses |